

Data Quality Assurance Framework (DQAF) & UN Fundamental Principles

GOVERNMENT OF INDIA

MINISTRY OF STATISTICS AND PROGRAMME IMPLEMENTATION (MoSPI)

NATIONAL STATISTICAL SYSTEMS TRAINING ACADEMY (NSSTA)

TRAINING HANDBOOK ON DATA QUALITY ASSURANCE FRAMEWORK (DQAF) & STATISTICAL GOVERNANCE

CHAPTER 1: PRINCIPLES, PRIVACY AND ACCREDITATION

1. MoSPI Data Quality Assurance Framework (DQAF)

The National Statistical Commission (NSC) of India mandates adherence to the Data Quality Assurance Framework (DQAF), structured around five primary dimensions:

- 1. Integrity and Professional Independence:** Statistics must be produced without political or administrative interference, ensuring impartiality, objectivity, and transparency in release schedules.
- 2. Methodological Soundness:** Adherence to international standards, including National Industrial Classification (NIC-2008), National Product Classification for Services (NPCS), and IMF Special Data Dissemination Standards (SDDS).
- 3. Accuracy and Reliability:** Rigorous survey design, sampling error estimation, validation via digital Computer Assisted Personal Interviewing (CAPI), and minimize non-sampling errors.
- 4. Serviceability and Timeliness:** Regular advance release calendars, periodic data revisions, and maintaining consistency across related statistical domains.
- 5. Accessibility and Clarity:** Dissemination through open data portals, accompanying metadata, microdata access protocols, and prompt statistical advisories.

2. UN Fundamental Principles of Official Statistics

Adopted by the UN General Assembly and formally approved by the Union Cabinet of India:

- Principle 1: Relevance, Impartiality, and Equal Access - Official statistics represent an indispensable public good.
- Principle 2: Professional Standards and Ethics - Scientific principles and professional ethics guide data collection methods.
- Principle 3: Accountability and Transparency - Clear metadata and methodological transparency for public interpretation.
- Principle 4: Prevention of Misuse - The statistical agency is entitled to comment on erroneous interpretation and misuse of statistics.

- Principle 5: Sources of Official Statistics - Choice of data source (surveys, administrative records) based on quality, cost, and respondent burden.
- Principle 6: Confidentiality - Microdata collected from natural or legal persons must be held strictly confidential and used exclusively for statistical purposes.
- Principle 7: Legislation - Laws and regulations governing statistical operations must be made public.
- Principle 8: National Coordination - Coordinated statistical activities across central ministries and State Directorates of Economics and Statistics (DES).
- Principle 9: International Standards - Adoption of international statistical concepts and classifications.
- Principle 10: International Cooperation - Bilateral and multilateral collaboration to improve national statistical systems.

3. Digital Personal Data Protection (DPDP) Act 2023 & Statistical Fiduciary

Under Section 17 of the DPDP Act 2023, personal data processed strictly for research, statistical, or archiving purposes is exempt from certain notice requirements, provided:

- a) The processing does not result in any significant decision or measure affecting the specific data principal (respondent).
- b) Adequate technical and organizational measures (de-identification, anonymization, k-anonymity, aggregation) are implemented to prevent re-identification.
- c) Microdata access to external researchers undergoes strict Disclosure Avoidance System (DAS) scrutiny.